BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 087 · Searching and Sorting Exam Practice

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How does merge sort combine two sorted lists?

Q2. What is algorithmic thinking?
A. Guessing a random order
B. Removing every input
C. Planning clear ordered steps
D. Drawing only screen layouts

Q3. “Compression is identical to encryption and primarily keeps data secret.” Correct this. Explain why.

Q4. An app asks for pupils' locations and contacts without need. An online service uses only one language. Correct both claims. Explain why. (a) Posting all personal records improves privacy. (b) Technology has no effect on culture or language.

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I Do · Read new knowledge

Key vocabulary

Linear search — Checks array elements sequentially.

Binary search — Repeatedly halves a sorted search range.

Bubble sort — Adjacent comparisons and swaps.

Insertion sort — Inserts keys into a growing sorted section.

Merge sort — Splits and merges sorted groups.

Sorted prerequisite — Requirement that data be ordered before binary search.

New learning

Choose a search method: Linear search inspects elements sequentially and works with unsorted lists. Binary search can discard halves but needs sorted input.

Trace before describing: When explaining an algorithm, show the actual array values, tested positions and changes—not merely the algorithm's name.

Three required sorting ideas: Bubble compares and swaps adjacent pairs; insertion places a key into a sorted portion; merge divides and combines sorted lists.

Answer what was asked: A question about one pass is not a request for complete sorting. A question about a search index uses zero-based array positions.

Worked example

A club has an unsorted list [5,2,9,1]. Explain which search method can be used immediately and trace the search for 9. [4 marks]

1. Because the list is unsorted, choose linear search [1].

2. Compare first item 5, not a match [1].

3. Compare second item 2, not a match [1].

4. Third item 9 matches, so return zero-based index 2 [1].

We Do · Practise with guidance

Guided 1

A sorted array is [2,4,6,8,10]. Trace binary search for 8. [4 marks]

Worked method: First mid index 2 = 6 [1]. Target 8 is larger [1], so search right indices 3–4 [1]. Mid = 3 value 8 found at index 3 [1].

Guided 2

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A pupil sorts [4,1,3] using bubble sort. Show the first complete pass. [3 marks]

Worked method: Compare 4 and 1 [1], swap to [1,4,3] [1]; compare 4 and 3, swap to [1,3,4] [1].

Guided 3

Explain why binary search should not be applied directly to [8,1,4,6] and how to prepare it. [3 marks]

Worked method: Array is unsorted [1]. Binary search discards halves based on order [1]. Sort ascending [1,4,6,8] before applying it [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the condition to print the position of 128 in this sorted array.

```
array drives = [32,64,128,256]
for i = 0 to 3
    if ... then
        print(i)
    endif
next i
```

write

Question 1

State the prerequisite for binary search. [1 mark] Source: Original OCR J277-style question; not an official past-paper original.

Question 2

Trace the first bubble-sort pass on [5,3,4] and state the output array. [4 marks] Source: Original OCR J277-style question; not an official past-paper original.

Question 3

A dataset [6,2,9,1] is unsorted. Explain how to search for 9 without sorting, then describe how insertion sort would sort the whole list. [6 marks] Source: Original OCR J277-style question; not an official past-paper original.

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Exit Ticket · Answer before checking

A binary search can safely discard half a list only when the relevant data is sorted.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Compare the next item in each list. Take the smaller. Repeat, then add any items left over.

Q2. C — Planning clear ordered steps

Q3. It reduces the storage space required for data, using a suitable lossless or lossy method.

Q4. (a) Sharing personal records can remove people's control over their data. (b) Technology can change language, access, traditions and who can take part. Consider benefits and harms, even when an action is legal.

I Do / We Do pages · Guided practice

Q1. First mid index 2 = 6 [1]. Target 8 is larger [1], so search right indices 3–4 [1]. Mid = 3 value 8 found at index 3 [1].

Q2. Compare 4 and 1 [1], swap to [1,4,3] [1]; compare 4 and 3, swap to [1,3,4] [1].

Q3. Array is unsorted [1]. Binary search discards halves based on order [1]. Sort ascending [1,4,6,8] before applying it [1].

You Do pages · Code First

```
array drives = [32,64,128,256]
for i = 0 to 3
    if drives[i] == 128 then
        print(i)
    endif
next i
```

Expected result: 2

You Do pages · Independent questions

Q1. The data must be sorted [1].

Q2. Compare 5>3 [1], swap to [3,5,4] [1]. Compare 5>4 [1], swap to [3,4,5] [1].

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Q3. Linear search is suitable for unsorted data [1], comparing 6, 2 then 9 [1], found at index 2 [1]. Insertion sort builds sorted section [1]: insert 2 before 6 [1], then 9 and finally 1 into the correct places, giving [1,2,6,9] [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: